

Alberto ROTA

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EXPERIENCE

<i>Present</i> FEB 2023	PhD Student Researcher - ASENSUS SURGICAL INC. Focus: Computer Vision Deep Learning methods for enhancing the spatial and contextual informative content of endoscopic image data, with focus on 3D reconstruction and occlusion restoration - Developed, applied and surpassed state-of-the-art models and pipelines targeted at recovering 3D information from 2D endoscopic image data, with strong focus on self-supervised frameworks [NDA] - Researched, developed and tested geometry-aware learned representations of 3D endoscopic spaces and 2D images [NDA] - Worked in structured teams, both in contributing and leading positions - Gained project management, time management and DevOps skills
<i>Present</i> SEP 2023	Teaching Assistant - NEARLAB MEDICAL ROBOTICS, POLITECNICO DI MILANO Primary Course: Technologies for Motor Behavior Analysis and Virtual Modeling Guest Lectures at: Medical Robotics and Technologies for Computer Aided Surgery Gained communication, didactic and public speaking skills
<i>Present</i> OCT 2023	Scientific Communicator - POLITECNICO DI MILANO Course: <i>Understanding Artificial Intelligence</i>  - Introduced the basics of neural networks and data-driven algorithms interactively to high school students, reaching out to over 16 schools and 500 students over the course of 2 years - Developed didactic and teaching skills

EDUCATION

<i>Ongoing</i> FEB 2023	Ph.D in Bioengineering - POLITECNICO DI MILANO & ASENSUS SURGICAL INC., MILAN, IT Focus: Computer Vision applications for enhanced spatial context awareness in surgical robotics
DEC 2025 JULY 2025	Visiting Research Fellow - CHAIR OF PHOTOGRAMMETRY AND REMOTE SENSING, TUM, MUNICH, DE Focus: Geometry-guided reconstruction for challenging visual domains.
DEC 2022 SEP 2020	MSc in Bioengineering - POLITECNICO DI MILANO, MILAN, IT Focus: AI and Computer Vision methods for 3D data in bioengineering; Virtualization of teleoperated surgical robotic environments

TECH STACK

<i>ML/AI</i>	Python, PyTorch, SciKit, WandB	<i>Research</i>	MATLAB, L ^A T _E X, Consensus
<i>CV</i>	OpenCV, Open3D, Huggingface, Rerun	<i>Robotics</i>	ROS, Unity
<i>DevOps</i>	Docker, Git, Slurm, Tailscale	<i>3D/CAD</i>	Blender, Autodesk Inventor
<i>Coding</i>	C, C#, C++	<i>Graphics</i>	Figma
<i>LLMs</i>	Claude, ChatGPT, Cursor, MCP	<i>Misc</i>	OpenFOAM, Wordpress, MS Office

AI TOOLING AND PROMPT ENGINEERING SKILLS

<i>Coding</i>	Advanced prompt engineering for code generation, debugging, refactoring, technical documentation, task decomposition and iterative development	<i>Claude, Cursor</i>
<i>Research</i>	Literature review automation, paper summarization, research synthesis, crafting of queries for comprehensive academic search and analysis.	<i>ChatGPT, Consensus</i>
<i>Writing</i>	Advanced prompt engineering for academic writing, manuscript revision, tone adaptation, clarity enhancement and communication.	<i>Claude, ChatGPT</i>
<i>Image Gen</i>	Competent in generating and refining synthetic visual content with diffusion-based models.	<i>DALL-E</i>

RESEARCH PROJECT CONTRIBUTIONS

- NOV 2025 **UnReflectAnything** - PROJECT LEAD AND PRIMARY CONTRIBUTOR
JUL 2025 Conceived, developed, and led the full research agenda of *UnReflectAnything*, an RGB-only framework that removes specular highlights from arbitrary images by predicting a highlight map and reconstructing a reflection-free diffuse image. [1] 
- Designed and implemented a physically grounded synthetic ground-truth pipeline for virtually rendering and compositing specular highlights onto any RGB image
 - Built and trained a token-space inpainting module capable of removing specularities and restoring diffuse illumination at the feature level
 - Achieved competitive or state-of-the-art performance across multiple quantitative metrics compared to leading delighting and reflection removal methods
- JUL 2025 **Self-Supervised Feature Matching in Endoscopic Surgery** - SOLE CONTRIBUTOR
DEC 2024 Developed an end-to-end self-supervised pipeline based on novel-view synthesis and contrastive optimization for semantic embedding adaptation of DINOv2 features towards a pixel matching task in the surgical endoscopy domain [2] [NDA]
- Ideated and developed an SSL pipeline for establishing pseudo-ground-matches in source-synthetic endoscopic image pairs to be used for contrastive learning
 - Trained an adapter for DINOv2 to produce localized semantics to be used for correspondance tasks
 - Surpassed state-of-the-art models pixel matching tasks on surgical endoscopy imagery
- JUL 2022 **μVES - microVascular Evaluation System** - LEADER AND PRIMARY CONTRIBUTOR
MAR 2020 Built a fully automated pipeline for the topological and morphological analysis of 3D micro-vascular networks images from confocal microscopy, with Deep-Learning-based confocal image segmentation and integration with a CFD simulation software [3] 
- Built and trained a 3D U-Net for segmentation of 3D confocal microscopy images.
 - Developed a complete pipeline for quantitative analysis inclusive of segmentation, skeletonization, and quantitative morphological measurements
 - Primarily contributed and lead a team of 4 researchers, mastering problem-solving and leadership skills
- DEC 2020 **STEVE - Surgical Training Enhanced Virtual Environment** - SOLE CONTRIBUTOR
FEB 2022 Built a virtual reality training environment targeting teleoperated surgical robotics, enhanced with visuo-haptic assistance-as-needed guidance, personalized adaptive difficulty and visual feedback for haptic force training [4] 
- Built a VR simulator for surgical robotics in C# with Unity, connected via ROS to a teleoperation console. Developed haptic assistance-as-needed guidance algorithms
 - Planned and conducted an experimental study for statistical validation of the effect of the guidance strategies
 - Supervised MSc students on the development and integration of surgical tasks with morpho-adaptive difficulty [5] and visual feedback for grasping force training
- MAR 2025 **OVIT - Ovarian Cancer Resectability Prediction** - MINOR CONTRIBUTOR
OCT 2024 Participated in the development of a Deep Learning decision-support-system for Ovarian Cancer treatment planning [6] 
- Contributed to the development, statistical validation and academic publication of the pipeline
 - Designed and assembled a heavy-duty multi-GPU workstation for the clinical Deep Learning workload, with remote SSH/VNC access, Dockerized environments, and strict adherence to GDPR and data privacy best practices

OPEN SOURCE CONTRIBUTIONS

- Maintainer* **Ground Control** - OPEN-SOURCE PYTHON PACKAGE
DEC 2024 A Terminal-based package for monitoring system hardware in real time with rich plots and graphics in the terminal. Aimed for multi-GPU machines and ML workflows.  & PyPI
- Maintainer* **DaSSHboard** - VS CODE EXTENSION
APR 2025 A stylish customizable VS Code extension to manage multiple SSH remote connections with a smart one-click dashboard for faster access to remote development.  & 

Contributor **MedView** - VSCode EXTENSION
JUL 2025 A fast environment to preview and interact with medical images directly from a code editor, with support for .nii and .dcm files, aimed at streamlining AI development tasks in the medical image analysis field.  & 

AWARDS

JUN 2023 **Best Application Award** - HAMLYN SURGICAL ROBOTICS CHALLENGE 2023
Haptic assistance for improving skill transfer in surgical robotics training 

APR 2022 **Best Development Award** - POLIMI CAPSTONE PROJECTS 2022
SPINTEST - Data-Driven Compliancy Assessment for Extra-Corporeal Centrifugal Blood Pumps 

SELECTED RESEARCH PAPERS

- [1] **Alberto Rota**, Mert Kiray, Mert Asim Karaoglu, Patrick Rukhamp, Elena De Momi, Nassir Navab, and Benjamin Busam. UnReflectAnything: RGB-only highlight removal by rendering synthetic specular supervision. *Submitted to CVPR 2026 [CONFIDENTIAL]* .
- [2] **Alberto Rota** and Elena De Momi. Self-supervised contrastive embedding adaptation for endoscopic image matching. *IEEE Transaction on Medical Imaging*, 2025 - *Under Review* .
- [3] **Alberto Rota**, Luca Possenti, Giovanni S Offeddu, Martina Senesi, Adelaide Stucchi, Irene Venturelli, Tiziana Rancati, Paolo Zunino, Roger D Kamm, and Maria Laura Costantino. A three-dimensional method for morphological analysis and flow velocity estimation in microvasculature on-a-chip. *Bioengineering & Translational Medicine*, 2023 .
- [4] **Alberto Rota**, Ke Fan, and Elena De Momi. Implementation and assessment of an augmented training curriculum for surgical robotics. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023 .
- [5] **Alberto Rota**, Federica Xianyi Sun, and Elena De Momi. Performance-driven tasks with adaptive difficulty for enhanced surgical robotics training. In *2023 IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, 2023 .
- [6] Francesca Fati, Marina Rosanu, Luigi De Vitis, and **Alberto Rota et al.** Deep learning for decision support in ovarian cancer treatment planning. *Nature Women's Health*, 2025 - *Under Review* .
- [7] Junling Fu, **Alberto Rota**, Shufei Li, Jianzhuang Zhao, Qingsheng Liu, Elisa Iovene, Giancarlo Ferrigno, and Elena De Momi. Recent advancements in augmented reality for robotic applications: A survey. In *MDPI Actuators*, 2023 .

DISCLOSURES

GDPR I authorize the processing of personal data according to EU Regulation 679/2016 or according to the reader's local regulations if not in the EU

Accessibility I authorize the publication and the complete accessibility of this CV according to the italian D. Lgs n. 33 of March 14 2013

NDA Research work in this CV tagged with [NDA] has been carried out under IP protection policies and a Non-Disclosure Agreement. Details available upon request and on a subject basis.